Name: Date:

Solve each problem. Show your work in the box.

Write your final answer on the answer line (or in the blanks shown).

Use the strategies you learned in Lesson 5.02.03.

1 Exponents and Operations

Evaluate each expression.

$2^3 + 5 =$ ______ | $3 \times 4^2 =$ ______ | $10^2 - 15 =$ ______

Answers

2 Parentheses with Exponents

Evaluate each expression.

$(2 + 3)^2 =$ ______ | $2^2 + 3^2 =$ ______ | $(5 - 1)^2 \times 2 =$ ______

Answers

3 Full Order of Operations

Evaluate step by step.

$3 + (4^2 - 6) \times 2 =$ ______ | $(8 - 3)^2 \div 5 + 1 =$ ______

Answers

4 Square Garden

Atlas plants a square garden with sides of 7 feet. He buys 3 extra plants for each square foot.

Write an expression using an exponent to find the total number of plants. Evaluate it.

Expression and Answer ______

5 Same or Different?

Sophia claims $(3 + 2)^2 = 3^2 + 2^2$ because "you can distribute the exponent just like multiplication."

Is Sophia correct? Evaluate both sides and explain why they are the same or different.

Explanation ______